

	(c)(ii) <u>Method 1</u> $V_{AB} = I_{AB} \times R_{AB}$ $9.6 = 2.4 \times R_{AB}$ $R_{AB} = 4.0 \, \Omega$ $\frac{1}{R_X} + \frac{1}{R} = \frac{1}{R_{AB}}$ $\frac{1}{R_X} + \frac{1}{6.0} = \frac{1}{4.0}$ $R_X = 12 \, \Omega$	<u>Method 2</u> $V_R = I_R \times R$ $9.6 = I_R \times 6.0$ $I_R = 1.6 \, A$ $I = I_R + I_X$ $2.4 = 1.6 + I_X$ $I_X = 0.8 \, A$ $V_X = I_X \times R_X$ $9.6 = 0.8 \times R_X$ $R_X = 12 \, \Omega$	C1 C1 A1
	(c)(iii) $P = I E$ $= 2.4 \times 24$ $= 57.6 \, W$		A1
	(d) With a constant e.m.f. and a decreasing current from battery OR With a constant e.m.f. and an increasing total resistance in the circuit the power provided by the battery decreases		A1 M0

4	(a)(i)		
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